THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 5 & 6: **Life Processes, Control and Coordination** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Rings of cartilage are present in the: 1

(a) alveoli **(b)** stomach **(c)** trachea **(d)** oesophagus

2. For the absorption of digested food, Student A says it occurs in the small intestine; Student B says in the stomach. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

3. In desert plants, stomata open at: 1

(a) noon **(b)** never **(c)** night **(d)** morning only

4. The main nitrogenous waste in humans is: 1

(a) ammonia **(b)** creatine **(c)** uric acid **(d)** urea

5. Which vessel carries blood from the heart to the body? 1

(a) aorta **(b)** pulmonary vein **(c)** vena cava **(d)** capillary

6. The growth of roots towards water is called: 1

(a) chemotropism **(b)** geotropism **(c)** phototropism **(d)** hydrotropism

7. For the hormone which promotes ripening of fruits, Student A says ethylene; Student B says auxin. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. Hypothalamus releasing factors stimulate the: 1

(a) pituitary **(b)** pancreas **(c)** testes **(d)** kidney

9. Dwarfism results from deficiency of: 1

(a) adrenaline **(b)** insulin **(c)** thyroxine **(d)** growth hormone

10. The sense of smell is detected by: 1

(a) olfactory receptors **(b)** photoreceptors

(c) gustatory receptors **(d)** thermoreceptors

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does the rate of breathing in aquatic organisms differ from terrestrial ones? 2

12. A student says capillaries have thick walls. Justify or refute. 2

13. What is the role of the receptor? Name the receptors for taste. 2

14. A student says that the cerebellum controls hunger. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What are the functions of the kidney? 3

16. Explain the feedback mechanism with reference to blood sugar. 3

17. Differentiate between arteries and veins (three points). 3

OR

What is the role of saliva in digestion? Where is it secreted? 3

18. How do nerve cells communicate with muscle cells? 3

OR

Why are hormones called chemical messengers? Give two examples of their role. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe transportation of water and minerals in plants. 5
(ii) What is root pressure?
(iii) When is it important?

OR

(i) Draw a schematic diagram of blood flow (describe) in the human heart and body. 5
(ii) What is meant by systole and diastole?

20. (i) Describe the parts and functions of the human brain. 5
(ii) What would happen if the cerebellum were damaged?

OR

(i) What are plant hormones? Name four with their functions. 5
(ii) How does a plant respond to the absence of light?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND